

Product: Catalog, Category Entities

- (i) A ProductStore links to one or more prodCatalogs.
- (ii) A prodCatalog contains multiple productCategories via ProductCategory.
- (iii) Each ProductCategory holds products via ProductCategoryMember.
- (iv) Category can be hierarchically nested via ProductCategoryRollup.

Catalog → collection / containers

Category → grouping / classification inside catalog.

Example: US Catalog
India Catalog
Winter Catalog
Marjama Catalog

Category → Mobile, shoes, Jefeetab

↓

↓ HP, Dell

iphone, Samsung

↳ (Products)

(1) ProdCatalog

prodCatalogId* → string (DemoCatalog)

CatalogName → Human-readable

useQuickAdd → Y/N

headerLogo

(2) ProductStoreCatalog
Store - Catalog Link

(Many-to-Many)

ProdCatalogId* → String (PK + FK) which catalog is being linked
ProductStoreId* → String (PK + FK) which store this catalog is attached
From Date* → PK

Product Store Id ProdCatalogId

A C1

A C2

B C1

(3) ProductCategory

Represent the grouping of products.
Categories can be hierarchical. Electronics → Laptop → Gaming.

ProductCategoryId* → String (Electronics, shoes, laptops)

ProductCategoryTypeId → FK (The type of this category)

PrimaryParentCategoryId → example → laptop → electronic (parent)

CategoryName → String ("Gaming Laptop")

description

CategoryImageURL → thumbnail image.

PrimaryParentCategoryId gives main parent. However, a category can belong to multiple parent category via ProductCategoryRollup - enabling cross-distinct.

→ example: "Gaming Laptop" appearing under both "Gaming" & "Laptop".

ProductCategoryRollup → ProductCategoryId*

parent ProductCategoryId*, From Date*

(4) ProductCategoryType

ProductCategoryTypeId → PK (unique type code)
ParentTypeId → FK
has table / description

CategoryType →	✓ Catalog - Category	→	Catalog
	Tax Category	→	Tax
	Usage - Category	→	Usage
	Material Category	→	Materials
	✓ Search - Category	→	Search

(5) ProdCatalogCategory (Catalog - Category - Mapping.)

ProdCatalogId → PK + FK
ProductCategoryId → PK + FK
ProdCatalogCategoryTypeId → PK + FK (One to this category)
Prom Date → PK
Exp Date → DateTime

ProdCatalogCategoryType ID :

- PCCT - Most Popular : Category showing most-ordered products.
- PCCT - Promotions : Category shown in the promotion section
- PCCT - Whats-new

(7) ProductCategoryMember Product in a Category.

ProductCategoryId *
ProductId *
Prom Date *

(8) ProductCategoryAttribute
Key - pair attribute that don't fit the standard fields.

productCategoryId*, attName*, attValue*

(9) ProductCategoryRole
Role for categories

productCategoryId*
roleId*
roleType* (owner, buyer)
joinDate*

(10) ProductCategoryContent
Content linked to category

productCategoryId* → PK + FK
contentId* → FK → Content (image, text, document)
productContentTypeId* → What type of content is this?
joinDate*
thruDate

productContentTypeId:

- Category-Image-URL
- Category-Image-ALT
- Category-description
- Category-name
- Link-one-image-URL

(1) ProductCatalogInventoryFacility
 = (Catalog Inventory Facility)

productCatalogId *
 facilityId *
 from Date *

Product Feature :

(2) ProductFeatureType

productFeatureTypeId *
 parentType Id

- Color - fabric
- brand - Model-year
- size - Hazmat
- Style - License

(3) ProductFeatureCategory

productFeatureCategoryId *
 parent Category Id
 description

(3) Product Feature

ProductFeatureId *
 productFeatureTypeId → size, Color, Brand
 productFeatureCategoryId →
 uomId → unit of measure (inch, kg, cm)
 defaultAmount → default price adjustment and when this
 feature is applied.

(4) ProductFeatureAppl
Feature Applied to product.

product Id * → The product
 productFeatureId * → feature being applied
 productFeatureApplTypeId * → Role of this feature on the product
 from date * → effective start date
 amount → one-time price adjustment for selecting this feature.

ApplicationTypes (productFeatureApplTypeId):

- Standard-Feature: Always included, no customer choice.
- Selectable-Feature: customer select from a list - eg, color, size.
- Required-Feature: customer must select this - cannot order without this.
- Optional-Feature: gift wrap.

(5) ProductFeatureInteractn

Feature Interaction (rules): Define rules between two features.

productFeatureId * → First Feature
 productFeatureId2 * → Second Feature
 productFeatureInteractnTypeId → Type of interaction

Interaction Type:

Feature-Interactn-Depend: Selecting feature A requires feature B.
 Feature-Interactn-Incomp: feature A cannot coexist with B.

(6) ProductFeatureGroup -

Group multiple features together for display or configuration.

ProductFeatureGroupId → unique group Id
 description → label shown to customer (Available "color")

(7) ProductFeatureGroupAppl

link a ProductFeature to a ProductFeatureGroup.

productFeatureGroupId *

productFeatureId *

from Date *

(8) ProductFeatureCategoryAppl

link a productCategory to a ProductFeatureCategory.

productCategoryId *

productFeatureCategoryId *

from Date *

(9) ProductFeatureCtnrAppl

links a productCategory to a ProductFeatureGroup.

productCategoryId *

productFeatureGroupId *

from Date *